MARYAM REHAN

INTERNSHIP REPORT CYBER SECURITY

Week 3 Task 7 — Malware Types & Common Attacks

Introduction

Cybersecurity threats are activities that can damage computer systems, steal sensitive information, or interrupt digital services. Malware and cyberattacks are common threats faced by individuals and organizations. A SOC (Security Operations Center) Analyst must understand these threats, recognize suspicious activities, and identify possible indicators of attacks.

Six Types of Malware

Malware is malicious software designed to damage systems, steal information, or gain unauthorized access.

Virus

A virus is malicious code that attaches itself to a legitimate file or program and spreads when the infected file is executed.

Worm

A worm is malware that can replicate itself and spread automatically across computers and networks without requiring a host file.

Trojan

A Trojan is malicious software disguised as a legitimate application to trick users into installing or executing it.

Ransomware

Ransomware is malware that encrypts files or blocks access to a system and demands payment to restore access.

Spyware

Spyware secretly monitors user activities and collects sensitive information such as passwords and browsing data.

Rootkit:

A rootkit is malicious software designed to hide its presence while maintaining unauthorized access to a computer system.

Common Cyberattacks

Phishing Attack

Phishing is a social engineering attack in which cybercriminals send fake emails, messages, or websites that appear to come from trusted sources. Their purpose is to trick users into sharing sensitive information, clicking malicious links, or downloading infected attachments.

How phishing works:

1. An attacker creates a fake email or message.
2. The message appears to come from a trusted organization.
3. The victim receives a suspicious link or attachment.
4. The victim clicks the link or enters personal information.
5. The attacker may steal credentials or install malware.

Detection Signs:

- Emails from suspicious or unknown senders.
- Fake website URLs and unusual links.
- Unexpected attachments.
- Urgent messages requesting passwords or personal information.

Denial of Service (DoS) and Distributed Denial of Service (DDoS)

A Denial of Service (DoS) attack attempts to make a website, server, or network unavailable by overwhelming it with excessive requests or traffic.

A Distributed Denial of Service (DDoS) attack uses multiple devices or systems to send large amounts of traffic toward a target simultaneously.

How DoS attacks work:

1. The attacker selects a target server or website.
2. A large number of requests are sent to the target.
3. The server becomes overloaded.
4. Legitimate users experience slow services or cannot access the website.

Detection Signs:

- Unusual increases in network traffic.
- Large numbers of incoming requests.
- Slow server response times.
- Firewall and network monitoring alerts.

Man-in-the-Middle (MITM) Attack

A Man-in-the-Middle attack occurs when an attacker secretly intercepts communication between two parties. The attacker may monitor or modify the information being exchanged without the users knowing.

How it works:

1. Two parties communicate over a network.
2. An attacker positions themselves between the communication.
3. The attacker intercepts the transmitted information.
4. Sensitive information may be monitored or modified.

Detection Signs:

- Unexpected security certificate warnings.
- Suspicious public Wi-Fi networks.
- Unusual network or ARP activity.
- Unexpected changes in communication sessions.

Brute Force Attack

A brute force attack is a password-guessing technique in which attackers repeatedly try different password combinations until they find the correct credentials.

How it works:

1. The attacker selects a user account or login page.
2. Automated tools attempt different password combinations.
3. The system receives repeated login attempts.
4. If a correct password is guessed, the attacker may gain unauthorized access.

Detection Signs:

- Multiple failed login attempts.
- Repeated authentication requests.
- Account lockout notifications.
- Successful login after several failed attempts.
- Suspicious login attempts from unfamiliar IP addresses.

How Ransomware Spreads

Ransomware is a dangerous type of malware that can spread through different methods:

1. Phishing Emails: Attackers send infected attachments or malicious links.
2. Malicious Downloads: Users unknowingly download infected software.
3. Software Vulnerabilities: Attackers exploit weaknesses in outdated applications or operating systems.
4. Compromised Credentials: Stolen usernames and passwords may allow attackers to access systems.
5. Network Sharing: Some ransomware can spread to connected computers and shared folders.

Impact of Ransomware

- Important files become encrypted or inaccessible.
- Organizations may experience business interruptions.
- Sensitive information may be stolen or exposed.
- Recovery and incident response can be expensive.
- Organizations may suffer reputational damage.
- Users may lose access to important personal or business data.

Detection Signs for Common Attacks

A SOC Analyst monitors system activities, network traffic, and security logs to identify suspicious behavior.

Phishing Attack: Suspicious emails, malicious links, unknown attachments, and emails from unfamiliar senders. A SOC Analyst can identify these signs by checking email and DNS logs.

DoS/DDoS Attack: A sudden increase in network traffic, unusually high numbers of requests, and slow server responses. These signs can be identified through firewall and network logs.

MITM Attack: Unexpected security certificate warnings, unusual ARP activity, and suspicious network communication. A SOC Analyst can investigate network and IDS logs.

Brute Force Attack: Multiple failed login attempts, repeated authentication requests, and suspicious login attempts from unfamiliar IP addresses. These activities can be detected through authentication and system logs.

Ransomware Attack: Unusual file encryption, mass file modifications, unexpected ransom notes, and inaccessible files. These signs can be investigated through endpoint and security logs.

Role of a SOC Analyst in Threat Detection

A SOC Analyst plays an important role in identifying and monitoring cybersecurity threats. Their responsibilities include:

1. Monitoring security alerts and system logs.
2. Identifying suspicious network activities.
3. Investigating malware infections and attack indicators.
4. Analyzing unusual login attempts.
5. Detecting phishing emails and malicious links.
6. Reporting security incidents and supporting incident response.

SOC Analysts use security tools such as SIEM (Security Information and Event Management), antivirus software, firewalls, and endpoint detection tools to monitor and investigate potential threats.

Preventive Security Measures

The following practices help reduce the risk of malware infections and cyberattacks:

- Use strong and unique passwords.
- Enable Multi-Factor Authentication (MFA).
- Avoid suspicious links and email attachments.
- Keep operating systems and applications updated.
- Use trusted antivirus and endpoint protection software.
- Create regular backups of important data.
- Avoid connecting to unknown or unsecured Wi-Fi networks.
- Monitor login activities and security alerts.

Learning Outcomes

After studying malware types and common cyberattacks, I learned:

1. The differences between six major types of malware.
2. How phishing attacks manipulate users.
3. How DoS and DDoS attacks affect network availability.
4. How MITM attacks intercept communication.
5. How brute force attacks attempt to guess passwords.
6. How ransomware spreads and affects individuals and organizations.
7. How SOC Analysts identify suspicious activities through logs and indicators.

Conclusion

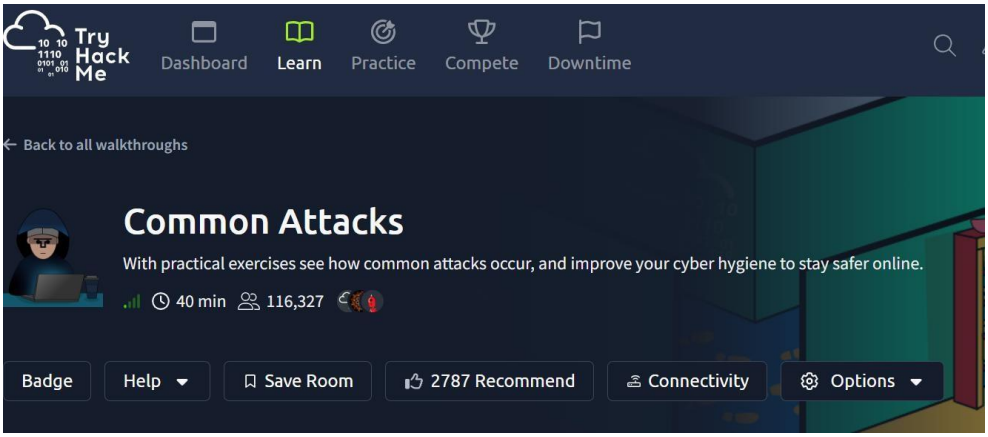
Understanding malware types and common cyberattacks is an important part of cybersecurity. Different attacks target systems, networks, and users in different ways. Learning their mechanisms and detection signs helps cybersecurity professionals identify potential threats and respond effectively.

This task improved my understanding of malware, common attack techniques, ransomware, and security monitoring. It also highlighted the importance of continuous monitoring, strong authentication, regular backups, and cybersecurity awareness in protecting digital systems and information.

A SOC Analyst can use system, network, firewall, authentication, and endpoint logs to identify suspicious activities.

1. **Phishing:** Suspicious URL requests and unusual email activity in email and DNS logs.
2. **DoS/DDoS:** A sudden increase in incoming traffic and repeated requests in firewall and network logs.
3. **MITM:** Unexpected ARP table changes and suspicious network communication.
4. **Brute Force:** Multiple failed login attempts recorded in authentication logs. Windows Event ID 4625 indicates a failed logon attempt.
5. **Ransomware:** Unusual file modifications, mass file encryption, and suspicious file access in endpoint logs.

TRY HACK ME ACTIVITY



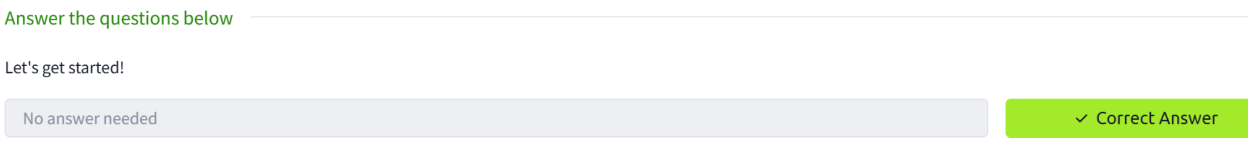




Objective

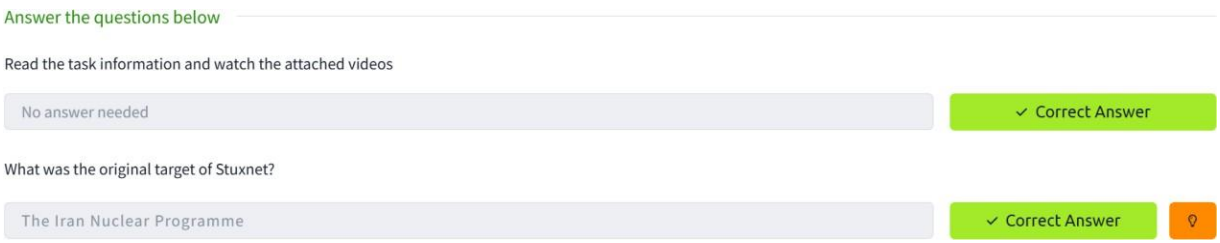
The objective of this task was to understand common cybersecurity attacks, how attackers target individuals and organizations, and the techniques they use to gain unauthorized access, steal sensitive information, or damage systems. It also aimed to develop awareness of social engineering, phishing, malware, ransomware, and password-based attacks, along with preventive security measures.

TASK 01



This task introduced the basic concepts of cybersecurity and explained why protecting digital information is important. I learned that cybersecurity helps maintain confidentiality, integrity, and availability of information. It also helps individuals and organizations protect their digital assets.

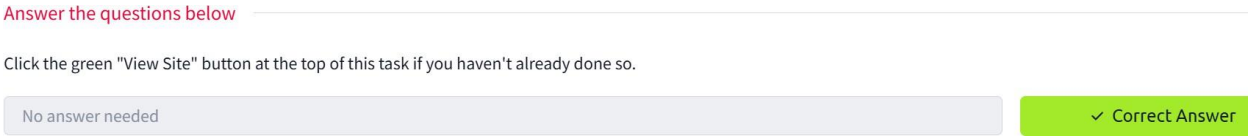
TASK 02



Social engineering is a technique used by attackers to manipulate people into revealing confidential information or performing unsafe actions. Attackers may pretend to be trusted individuals, employees, or organizations.

I learned that social engineering attacks often depend on human emotions such as fear, urgency, curiosity, and trust. Users should verify requests before sharing sensitive information.

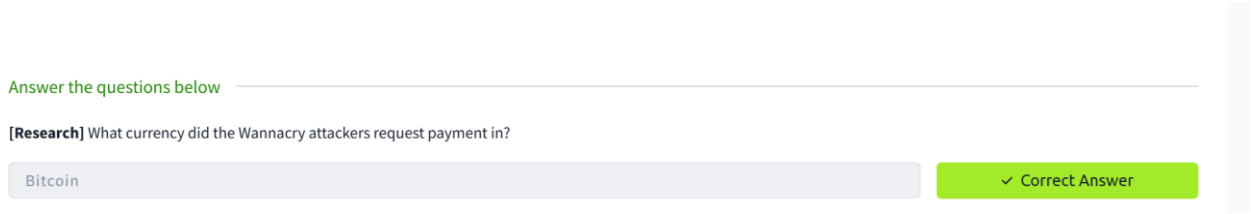
TASK 03



Phishing is a cyberattack in which attackers send fake emails, messages, or websites to steal personal information, passwords, and financial details.

I learned how to identify phishing attempts by checking suspicious links, unknown senders, spelling mistakes, urgent messages, and unexpected attachments. Avoiding suspicious links and verifying website addresses can help prevent phishing attacks.

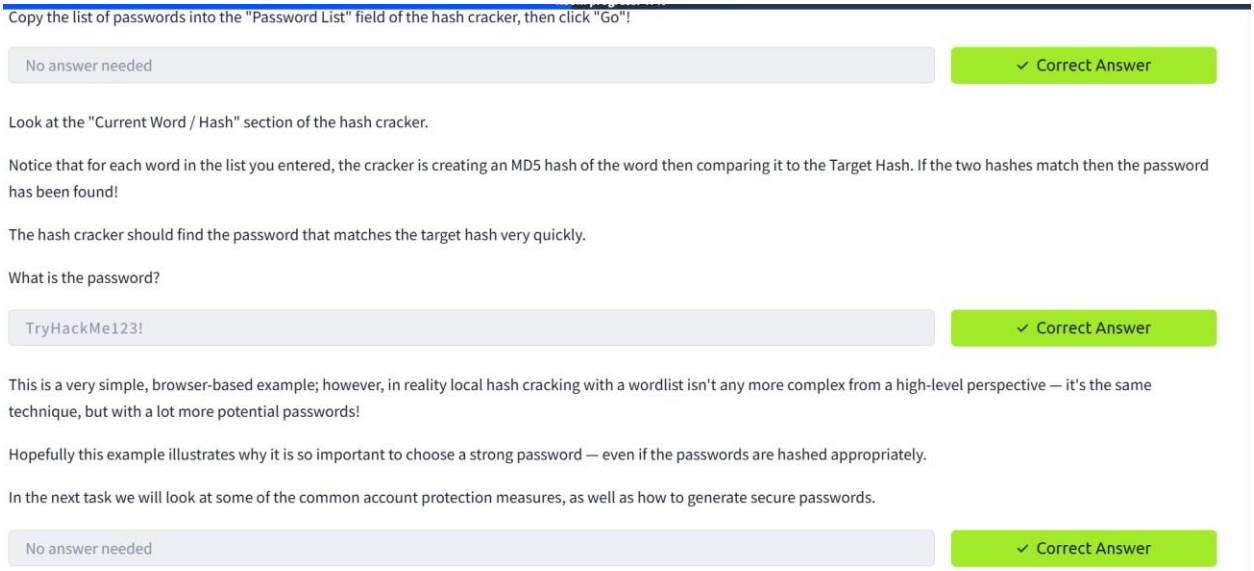
TASK 04



Malware is malicious software designed to damage systems, steal information, or gain unauthorized access. Common types include viruses, worms, Trojans, spyware, and ransomware.

Ransomware is a type of malware that encrypts files and demands payment to restore access. I learned the importance of using antivirus software, avoiding unknown downloads, and keeping regular backups.

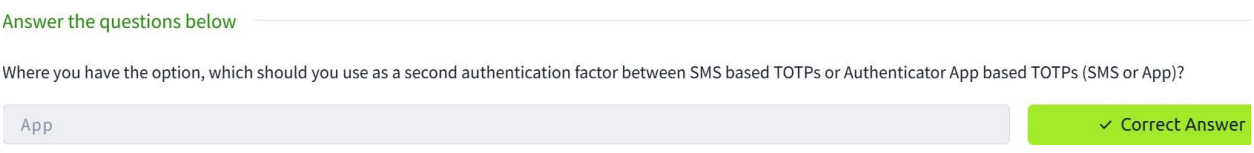
TASK 05



Passwords are an important security measure used to protect online accounts and digital systems. Weak or reused passwords make accounts vulnerable to cyberattacks.

I learned that strong passwords should be long, unique, and difficult to guess. Password managers can help securely store passwords, while multi-factor authentication provides an additional layer of protection.

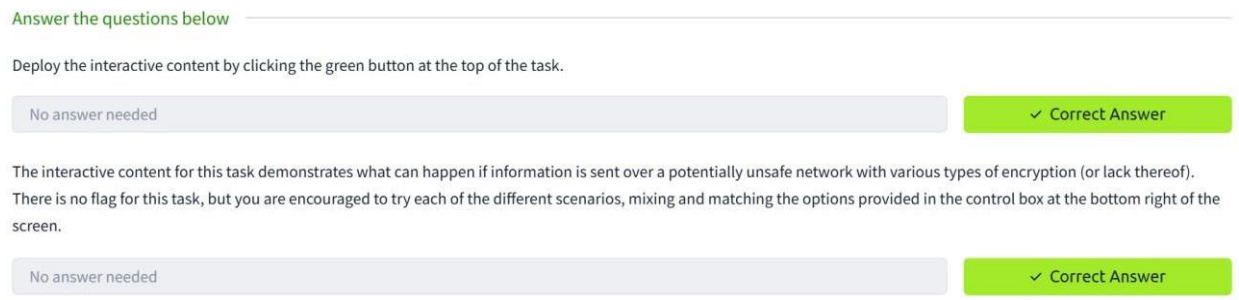
TASK 06



Multi-Factor Authentication (MFA) adds extra security by requiring more than one method of verification during login. These factors may include something you know, something you have, or something you are.

I learned that authenticator applications provide time-based one-time passwords (TOTPs), which are generally safer than SMS-based codes. Password managers also help generate and store strong, unique passwords.

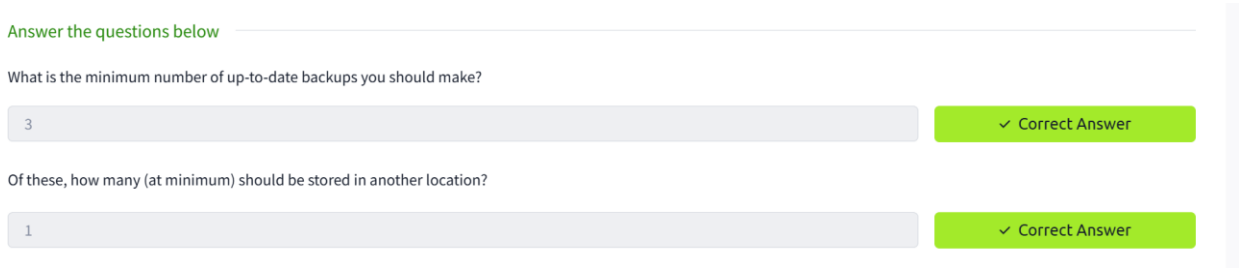
TASK 07



Public Wi-Fi networks are commonly available in cafes, airports, hotels, and other public places. However, unsecured public networks can expose users to security risks.

I learned that users should avoid accessing sensitive accounts on unsecured networks, use HTTPS websites, enable a trusted VPN when appropriate, and disable automatic Wi-Fi connections.

TASK 08



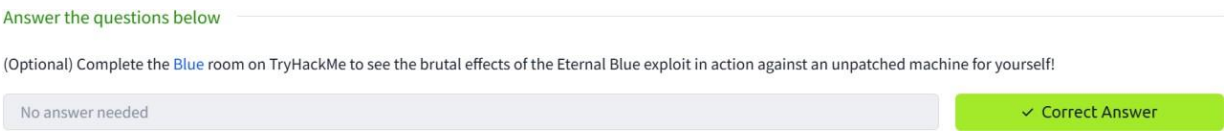
Backups are copies of important data that help recover information after accidental deletion, hardware failure, malware infection, or ransomware attacks.

I learned about the 3-2-1 backup rule:

- Keep at least 3 copies of important data.
- Store copies on at least 2 different storage mediums.
- Keep at least 1 backup off-site.

Regular backups help reduce the risk of permanent data loss.

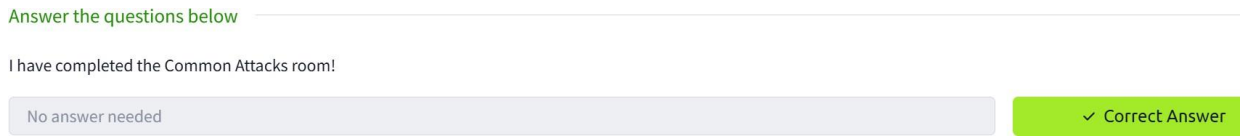
TASK 09



Software updates and security patches fix vulnerabilities, improve performance, and protect devices against newly discovered threats.

I learned that outdated operating systems, applications, and software can create security risks. Enabling automatic updates and installing security patches regularly helps maintain system security.

TASK 10



This task summarized the importance of cybersecurity awareness and responsible online behavior. I learned that understanding cyber threats and following security best practices can significantly reduce the risk of cyberattacks.

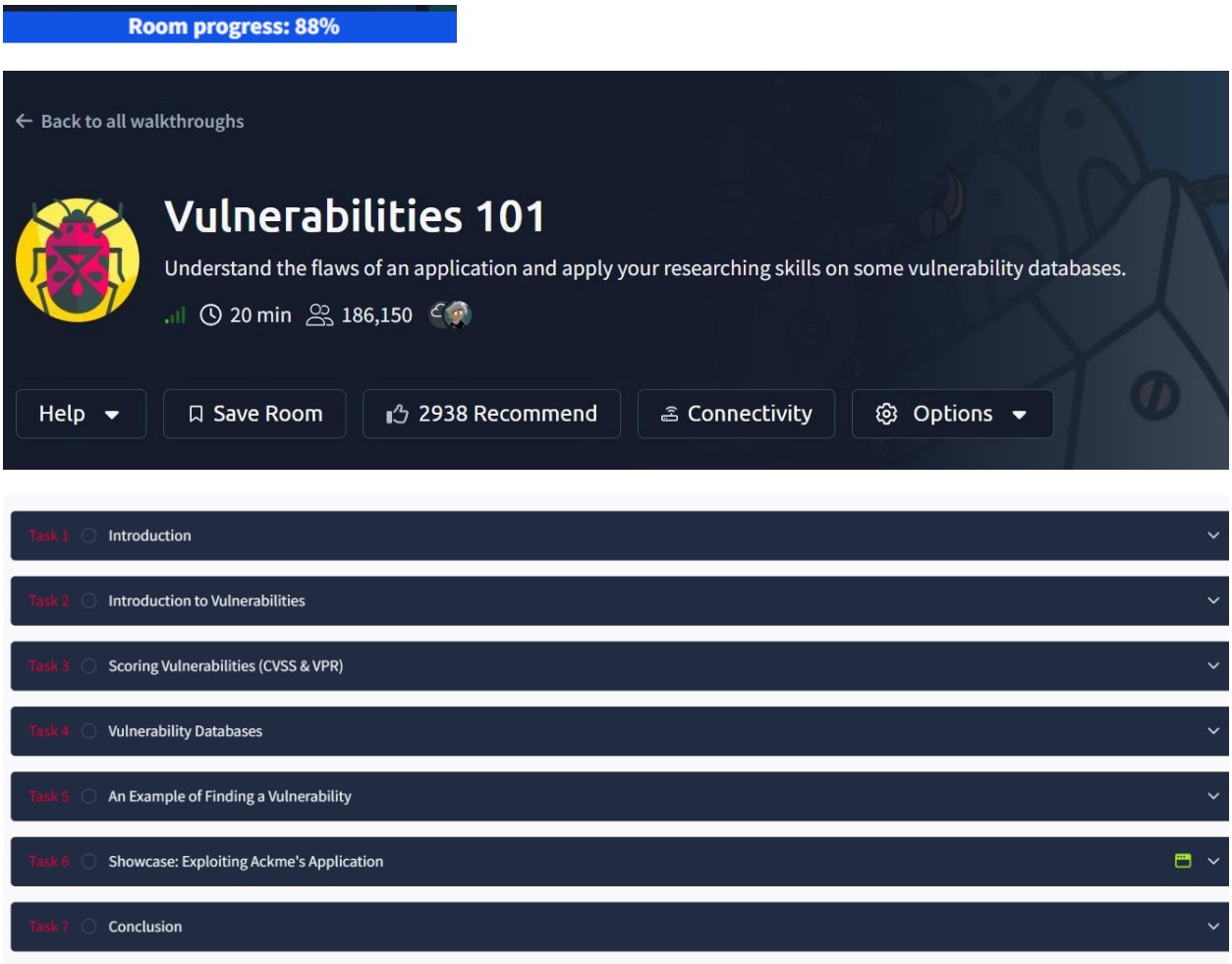
Overall Understanding

Through these ten tasks, I developed an understanding of common cybersecurity threats and preventive security practices. The main lessons learned include:

1. Recognizing and avoiding social engineering and phishing attacks.
2. Understanding malware and ransomware risks.
3. Creating strong and unique passwords.
4. Using MFA and password managers.
5. Staying safe while using public Wi-Fi.
6. Following the 3-2-1 backup rule.
7. Installing software updates and security patches regularly.

Conclusion

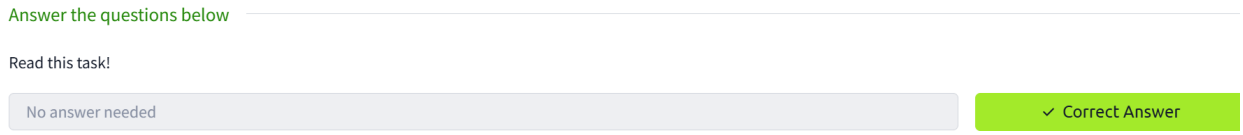
Cybersecurity is everyone's responsibility. This learning activity helped me understand how cybercriminals target individuals and organizations and how simple security practices can prevent many attacks. I now understand the importance of strong authentication, secure browsing, regular backups, software updates, and awareness of suspicious online activities. By applying these practices in daily life, individuals can improve their digital safety and protect their personal information.



Objective

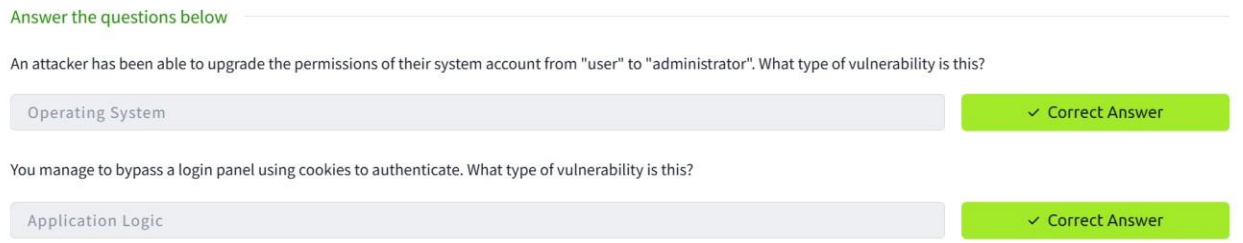
The objective of this learning activity was to understand the fundamentals of cybersecurity vulnerabilities, their identification, assessment, and exploitation. The room aimed to introduce vulnerability scoring frameworks, vulnerability databases, and basic penetration testing techniques. It also provided practical knowledge of how security weaknesses can be discovered and evaluated to improve system security.

TASK 01



In this task, I learned the basic concepts of cybersecurity vulnerabilities and why vulnerability management is important. I understood that vulnerabilities are weaknesses in computer systems, networks, or applications that attackers may exploit to gain unauthorized access or perform malicious activities. I also learned that identifying security weaknesses is an important part of protecting digital information and infrastructure.

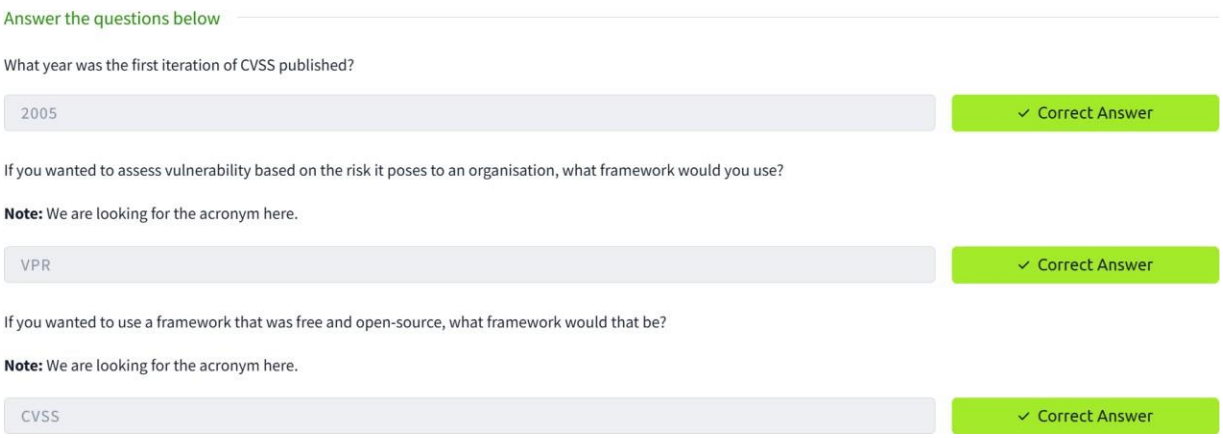
TASK 02



In this task, I learned about the meaning and different categories of vulnerabilities. A vulnerability is a weakness or flaw in the design, implementation, or configuration of a system. I studied five main categories:

- Operating System: Weaknesses in operating systems that may allow privilege escalation.
- Misconfiguration: Incorrectly configured applications or services.
- Weak or Default Credentials: Easily guessed or unchanged default passwords.
- Application Logic: Errors in application design or authentication mechanisms.
- Human Factor: Weaknesses caused by human behaviour, such as phishing.

TASK 03

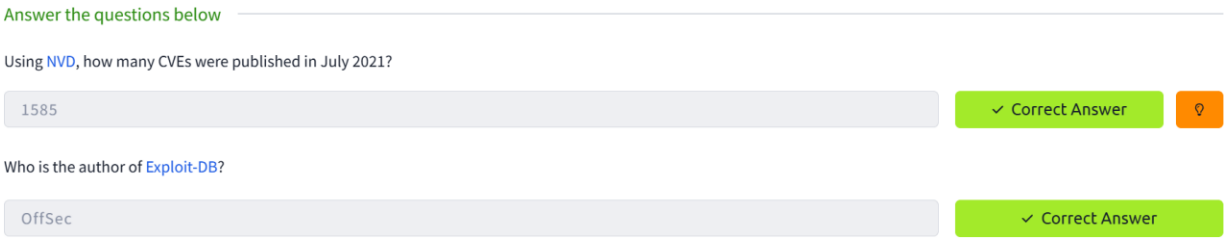


In this task, I learned how organizations evaluate and prioritize vulnerabilities according to their severity and risk. I studied two important frameworks:

- Common Vulnerability Scoring System (CVSS): Used to measure the severity of vulnerabilities based on factors such as exploitability and impact on confidentiality, integrity, and availability.
- Vulnerability Priority Rating (VPR): Focuses on the actual risk a vulnerability poses to an organization and helps prioritize security fixes.

I also learned about vulnerability severity levels, including Low, Medium, High, and Critical.

TASK 04

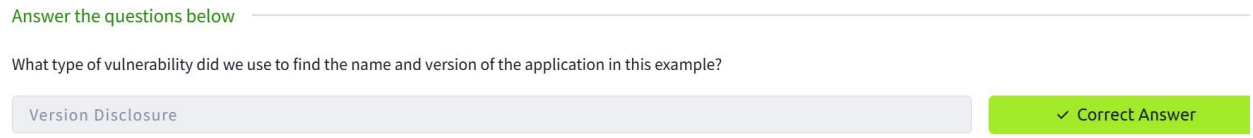


In this task, I explored two important online resources used by cybersecurity professionals to research known vulnerabilities.

- National Vulnerability Database (NVD): Provides information about publicly known vulnerabilities, including their CVE identification numbers and severity details.
- Exploit-DB: Contains publicly available exploit information and Proof of Concept (PoC) code for different applications and software.

I learned the difference between vulnerabilities, exploits, and Proof of Concepts and how these databases support vulnerability research.

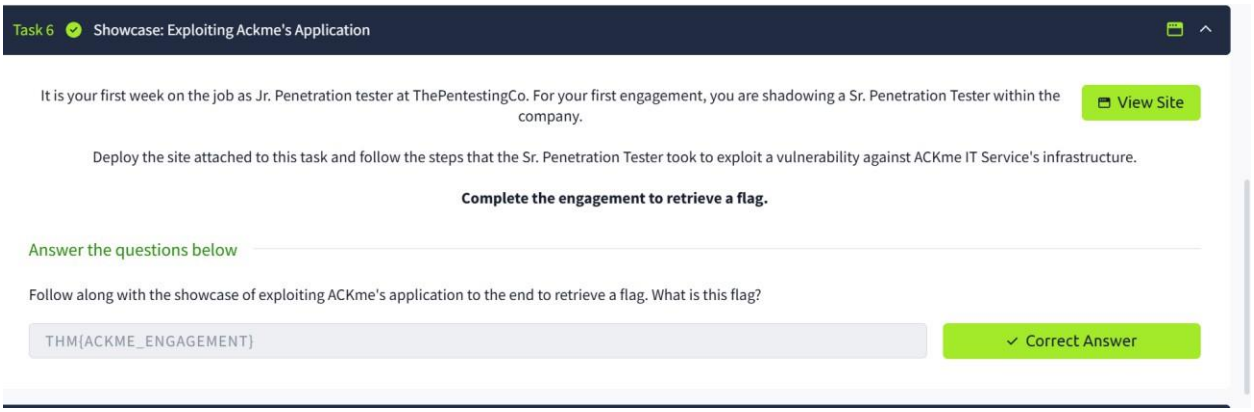
TASK 05



In this task, I learned how a small information disclosure can help identify more serious security weaknesses. The example demonstrated how identifying an application's name and version can help security researchers search for related vulnerabilities in Exploit-DB.

I understood that version disclosure can provide useful information during security assessments. By researching the exact application version, penetration testers can identify potentially relevant exploits and understand the security risks associated with outdated software.

TASK 06



In this task, I followed a practical penetration testing demonstration involving ACKme IT Services. I learned how to perform initial reconnaissance and identify services running on a target system.

The main activities included:

1. Reviewed the engagement instructions and identified the authorized target IP address.
2. Used Nmap to scan the target system.
3. Identified three open ports: 22 (SSH), 80 (HTTP), and 443 (HTTPS).
4. Discovered the application's version number, 1.5.2.
5. Explored the vulnerability information and followed the provided exploitation demonstration.
6. Successfully obtained administrator-level access to the ACKme system.

7. Retrieved the final TryHackMe flag.

TASK 07



In this final task, I reviewed the important concepts covered throughout the Vulnerabilities 101 room. I improved my understanding of identifying vulnerabilities, evaluating their severity, researching known security weaknesses, and understanding the exploitation process.

I also learned the importance of vulnerability databases such as NVD and Exploit-DB and assessment frameworks such as CVSS and VPR. The practical ACKme demonstration provided an opportunity to understand how vulnerability research and penetration testing techniques work together.

Overall, this learning experience improved my basic cybersecurity knowledge and introduced me to important vulnerability assessment and penetration testing practices.

Overall Understanding

Through this room, I learned about different types of vulnerabilities, including operating system vulnerabilities, misconfigurations, weak credentials, application logic flaws, and human-factor vulnerabilities.

I also understood the importance of vulnerability scoring frameworks such as CVSS and VPR for evaluating and prioritizing security risks. I explored NVD and Exploit-DB to research publicly known vulnerabilities, CVEs, and available exploits.

During the practical ACKme IT Services demonstration, I performed an Nmap scan, identified open ports (22, 80, and 443), discovered application version 1.5.2, and followed the exploitation process to obtain administrator-level access.

Retrieved Flag: THM{ACKME_ENGAGEMENT}

This activity helped me understand how reconnaissance, vulnerability research, assessment, and exploitation are connected in penetration testing.

Conclusion

In conclusion, this learning activity improved my understanding of vulnerability management and basic penetration testing concepts. I learned how to identify security weaknesses, evaluate their severity, and use vulnerability databases to research potential threats.

The practical ACK me demonstration helped me understand the importance of identifying application versions and investigating possible vulnerabilities. Overall, this room strengthened my foundational cybersecurity knowledge and highlighted the importance of regularly identifying and fixing vulnerabilities to protect systems, applications, and sensitive information.

Room completed: 100%